



**GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY,
ODISHA, GUNUPUR**

BEST PRACTICE – 1



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TITLE OF THE PRACTICE: Roof Top Solar Energy

OBJECTIVE:

- Solar energy is a renewable resource, as it can be harnessed to generate electricity as long as the sun exists. India plays a significant role in the global education sector.
- The gross enrollment ratio (GER) for higher education in India is 28.4% for the 18-23 age group, providing an ideal opportunity for installing solar rooftops in educational institutions.
- This not only supports the use of renewable energy but also helps reduce energy expenses. Establishing a solar photovoltaic (PV) power plant can lower overall energy costs for the facility while promoting a greener, more sustainable environment.

OBJECTIVES OF THE PRACTICE:

- Install solar power plants on the rooftops of various university buildings.
- Reduce greenhouse gas emissions.
- Utilize the generated solar energy as a replacement for conventional electricity in the university.
- Export any surplus solar energy to the national grid.
- Provide students with practical experience and knowledge in solar energy systems.
- Promote long-term research in the field of solar power.
- Set an example for the public by adopting eco-friendly, green initiatives.

CONTEXT:

In addition to implementing an effective teaching-learning process, the Institute is committed to long-term sustainability. Our Institute has installed a solar energy system on the roof, which significantly meets its electricity needs. The installation of this system serves as a backup during power outages, leading to savings of around 30% since its installation.

THE PRACTICE:

University has installed 1023 number of Solar panels. Approximately 6,69,271 Units of solar electrical energy is generated per year and wheeled on the grid. The solar panels are mounted on the rooftop of MBA Block, BSH Block, Central mess and in Hostels (NC-10,11 & 13). The solar energy is effectively used in the Fountain system in the campus.

A Solar Electric Vehicle is used inside the campus to reduce the greenhouse gas emission. Apart from this we have conducted various workshops and training programs to develop future solar technicians. A one-week training program was conducted for ITDA Gunupur staff.

Two workshops were conducted to make the students understand the basic concepts involved in solar PV systems, wind farms etc. and to facilitate the research and development activities in this area.

EVIDENCE OF SUCCESS:

The evidence of success for a rooftop solar are stated as follows.

- Electricity Bill Reduction: To quantify savings, the electricity bills were reduced by 30% after installation.



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- Return on Investment (ROI): The payback period of total investment by the annual savings is around 4.5 years.
- Reduced carbon footprint: The university has kept in efforts to reduce carbon footprints.

PROBLEMS ENCOUNTERED AND RESOURCES REQUIRED

- **Weather Dependent:** There is a chance that the sky may be cloudy or rainy during the day, resulting in minimal or no sunlight. This reduces the reliability of solar panels as an energy solution.
- **Skilled Labor:** A skilled workforce is needed for maintenance and cleaning. As a result, operational costs will rise.
- **High upfront cost:** Solar panels require inverters and storage batteries to produce power, and while panel installation is relatively inexpensive, the additional equipment is often costly.
- **High surface area needed:** Solar power plants require large land areas, which cannot be used for other purposes while occupied by the panels.
- **Service:** Poor service from suppliers after installation.

IMPACT OF THE PRACTICE:

Reduction in Carbon Footprint: Decreases the institution's reliance on fossil fuels, leading to lower carbon emissions and a positive environmental impact.

Cost Savings: Significant reduction in energy bills by utilizing free solar power, freeing up resources for other institutional needs.

Energy Independence: Enhances energy security by reducing dependence on external power sources and price fluctuations.

Hands-on Research and Innovation: Provides a platform for faculty and students to engage in real-world renewable energy research and innovation.

Contributes to National Clean Energy Goals: Aligns with government initiatives to increase the share of renewable energy in the national grid.

Increased Institutional Reputation: Positions the university as a leader in sustainability, enhancing its reputation and attracting environmentally conscious students and partners.

Encourages Eco-friendly Practices: Promotes awareness and a culture of sustainability among students, staff, and the local community

NOTES (OPTIONAL)

Data Analysis Tools:

- Utilizing tools like Excel, Python, or specialized solar performance software to analyze data and generate reports.
- The evidence of success will be recorded by systematically tracking these metrics and maintaining thorough documentation.

Rewards:

- Energy Conservation Award.



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- All India level Green Building-1st prize with a cash prize of 3.5 Lakhs.
- Incubation support of 50 thousand to develop a solar micro grid kit.

ROOF TOP SOLAR ENERGY

The arial view of solar panel installation on Central Mess covering the entire roof. The solar panels are uniformly arranged, indicating a well-organized solar power system. Surrounding the building named NC – 2 also has roof top solar panels.

Solar panels on roof top of Central Mess (Drone).



Solar Panels on roof top of Knowledge Centre





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Solar Panels on roof top of BSH



Roof Top Solar Panels on Central Mess



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REPORT ON GENERATION OF POWER ON SOLAR POWER Dated : From 9th July to 31st August 2024

Year	Solar Power Phase – 1 (125KWP)	Solar Power Phase – 2 (375 KWP)
2019	49,267 Units	NA
2020	128,563 Units	NA
2021	134999 Units	NA
2022	1,17644 Units	NA
2023	133,048 Units	NA
2024	59,983 Units	46167 Units
Total	623104 Units	46167 Units



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Figure: Roof Top Solar Panels on Knowledge Centre building.